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Reinforcement Learning Researcher

Neurosymbolic RL · Planning · LLM Embodied Agents · Safe AI

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ML researcher (Ph.D. candidate, NTU Singapore) with 5+ years building **deep RL** and **planning** systems for control, navigation and energy: **PPO, SAC, A2C, distributional RL**, **hierarchical RL**, **MCTS**, **LLM-based planning for embodied agents** and **RL fine-tuning of LLMs (GRPO)**, in **PyTorch**; hands-on robotics with **ROS 2** on **Jetson AGX** (F1Tenth race car). Core focus: **neurosymbolic RL**, combining neural learning with symbolic planning, programmatic policies and formal verification to build agents that are **sample-efficient, robust, interpretable and verifiably safe**. Former ML Engineer on an RL project with **Google X**; first-author papers at **JMLR, ECML-PKDD, ECAI** and **IEEE ICWS**.

Technical Skills

Reinforcement Learning — PPO, SAC, A2C, QR-DQN / distributional RL, option-critic & hierarchical RL, goal-conditioned RL, reward shaping, sparse-reward & long-horizon tasks, robust & safe RL, neurosymbolic & programmatic policies, world models/MDP latent models (WAE-MDP)

Planning & Search — Monte Carlo Tree Search (MCTS), belief-space planning, D* / A* replanning, symbolic task planners, **LLM-based planning for embodied agents**

LLMs & Deep Learning — RL fine-tuning of LLMs (GRPO, RLHF, DPO), transformers from scratch, LLM steering for safe embodied planning, LLM-generated policies, Decision Transformers, mechanistic interpretability, MoE-style Top-k routing, variational encoders, differentiable decision trees, computer vision

Verification — Formal verification of RL controllers, reachability analysis, abstract interpretation, LP/MILP (Gurobi)

Tools — Python, PyTorch, Hugging Face (Transformers, TRL), Stable-Baselines3, CleanRL, Gymnasium / OpenAI Gym, MuJoCo, MiniGrid, ROS 2, SLAM & localisation (EKF, particle filter), NVIDIA Jetson AGX, MATLAB, Arduino, LaTeX

Experience

2023 – Present — **Ph.D. Researcher, Safe & Efficient RL** — Singapore
Nanyang Technological University (NTU) & CNRS@CREATE
- Led 5 end-to-end RL research systems (hierarchical RL, MCTS planning, distributional RL, formal verification); 2 open-sourced.

2022 – 2023 — **Research Associate** — Singapore
CNRS@CREATE
- Built a **deep reinforcement learning controller** for fine-grained, real-time energy management of a grid-connected **microgrid**.
- Mitigated forecast uncertainty in real-time operation (*Sustainable Energy, Grids and Networks*, Elsevier 2024); extended to interpretable programmatic RL policies (AI4X 2025).

2021 – 2022 — **Machine Learning Engineer** — Pune, India
InFocusp Innovations, on attachment to Google X (USA)
- Developed **reinforcement learning** agents for the **Abstraction and Reasoning Corpus (ARC)**, a benchmark for abstract reasoning and program synthesis, as part of a Google X project.

2021 — **Research Engineer, Autonomous Vehicles** — Hyderabad, India
TiHAN, IIT Hyderabad
- Built drive-by-wire **braking and throttle control** for an autonomous test vehicle on Arduino.
- Deployed a **lane-detection** deep learning model on the **NVIDIA Jetson AGX** edge platform.

Selected Projects

ECML-PKDD 2026 published — **Neurosymbolic Hierarchical RL with MCTS Planning** — code
STACK PyTorch · MuJoCo AntMaze · Point-Maze · MCTS · D* · goal-conditioned RL
- Two-level agent: a **symbolic planner** that updates its world model online, driving **goal-conditioned neural policies**.
- Designed **Belief World Tree Search**, an **MCTS** variant for planning over uncertain maps, with a convergence proof.
- **30–100× fewer samples** and **20× faster** than the SOTA HRL baseline; **25% fewer steps** than D* on MuJoCo AntMaze.

2025 – 2026 personal

RL Fine-Tuning of Language Models with GRPO

code

STACK PyTorch · Hugging Face Transformers · TRL · GRPO · GSM8K

- Implemented **GRPO from scratch** (group-relative advantages, PPO clipping, KL penalty to a reference model) to teach a from-scratch **GPT**, pretrained only on addition, to perform **subtraction**.
- Applied the same recipe to a pretrained **LLM** (SmolLM2-360M-Instruct) with **TRL** on GSM8K math reasoning; open-sourced as a tutorial series covering **RLHF, DPO and GRPO**.

2025 – 2026 real hardware

F1Tenth Autonomous Race Car: Full ROS 2 Navigation Stack

code

STACK ROS 2 · Python/C++ · NVIDIA Jetson AGX · LiDAR · RealSense IMU · VESC · SLAM Toolbox · Flask

- Coded and deployed the race stack on the car's **Jetson AGX**: **SLAM** mapping, **EKF** odometry–IMU fusion, **particle-filter** localisation, and **Pure Pursuit** trajectory tracking.
- Wrote a LiDAR **time-to-collision emergency-braking** safety node, a gap-following planner, and a Flask **mission-control dashboard** for launch, teleop and odometry tuning.

JMLR 2026
ECAI 2024
published

Differentiable Decision-Tree Policies for Interpretable Deep RL

code

STACK PyTorch · PPO · SAC · Stable-Baselines3 · CleanRL · Gymnasium

- Created **DTSemNet**, an exact neural encoding of decision trees trained by **plain gradient descent**; a **drop-in policy network** for RL pipelines.
- Trained with **PPO** and a modified **SAC** on continuous-control tasks (BipedalWalker, LunarLander), **matching neural policies** while staying human-readable.
- **Annealed Top-k routing** (MoE-style): beat the best baseline on 80% of regression and all classification tasks.

AAMAS 2027
under review

Robust Hierarchical RL via Distributional Critics (CriOC)

STACK PyTorch · PPO · A2C · QR-DQN · option-critic · MiniGrid · Gymnasium

- **Distributional option-critic** agent: quantile critics estimate state criticality; a **variational encoder** spreads risk evenly across sub-policies.
- PPO/A2C variants on 9 benchmarks: **+44% worst-case return** under adversarial perturbations (−12% nominal).

ICLR 2027
under review

Formal Safety Verification of RL Controllers

STACK PyTorch · Gurobi LP/MILP · abstract interpretation · reachability analysis

- Learn-then-verify pipeline that **formally certifies reach-avoid and safety** of RL policies via LP/MILP.
- **11–98× smaller certificates** than axis-aligned baselines, incl. a quadrotor controller.

IEEE ICWS 2021
ICSOC 2020/21
published

Deep RL for Resource Allocation in Edge Computing

STACK Python · deep RL · combinatorial optimisation

- **Deep RL agent for user allocation** and service placement under QoS and workload fluctuations.

Technical Leadership & Mentoring

Ongoing

SafeDAG: Precedence-Constrained LLM Steering for Safer Planning in Embodied Agents

Final-year project supervisor, NTU (ongoing)

- Steering **LLM planners** with precedence (DAG) constraints so embodied agents produce safe, correctly ordered plans.

2023 – Present

Student Projects & Teams

NTU final-year projects, URECA undergraduate research, F1Tenth

- Supervised **LLM-generated interpretable RL policies**, **ARC with LLMs**, **Decision Transformers**, mechanistic interpretability and CNN robustness.
- Technical lead of the **F1Tenth autonomous racing team**: built the race stack (above) and guided students on control and navigation; VP of CCDS student club (Deep Learning Bootcamp, AI Meetup).

Education

2023 – 2027 expected

Ph.D., Computer Science · Nanyang Technological University (NTU) — Singapore

Thesis: *Neurosymbolic Techniques for Safe and Efficient RL*. Advisors: Arvind Easwaran (NTU), Blaise Genest (CNRS).

2019 – 2021

M.Tech., Computer Science · Indian Statistical Institute (ISI) — Kolkata, India

Thesis: *User allocation & service placement in edge computing*. Advisor: Ansuman Banerjee. CPI 82.1%; top-5 thesis nomination.

2012 – 2016

B.Tech., Electronics & Instrumentation · NIT Silchar — Silchar, India

Thesis: *Real-time DC motor position control with PID and fuzzy logic*. Advisor: Munmun Khanra.

Awards & Publications

Awards DesCartes Ph.D. Scholarship, CNRS@CREATE (2023) · M.Tech. Scholarship, ISI Kolkata (2019).

Publications 5 first-author peer-reviewed papers: **JMLR** (2026), **ECML-PKDD** (2026), **ECAI** (2024), **SEGAN** (Elsevier, 2024), **IEEE ICWS** (2021); plus 2 at ICSOC (2020, 2021) and 3 under review (ICLR, AAMAS, AAAI).